Too Big to Vote On: Why Polymarket's Oracle Cannot Protect Its Largest Markets

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***Abstract*—What stops the voters who settle a prediction market from buying the unlikely answer cheaply and then voting it in? Polymarket settles disputed questions by a stake-weighted vote of UMA holders. Its designers answer with the stake: a coalition able to force an answer would own most of it, and the stake would lose value once the manipulation showed. That answer fails for the largest markets, because the prize grows while the vote is held. Shares are created in pairs backed by one dollar, so buying the expected answer also creates a share of the other. That share sells for a tenth of a cent and pays a dollar if the vote is captured. In contested markets, 91% of what these markets owed when the vote ended was created after the vote was called. In the largest, the prize reached 7.5 times the value of the staked supply, and on forks of both chains the deployed contracts pay a captured vote in full once a permissioned resolver settles the request. We prove that no rule that counts votes protects a market worth more than what a visible manipulation would cost the stake that votes. Halting trading during the vote caps the prize at its size when the vote was called, which the operator can weigh against the stake.**

1. Introduction

A prediction market sells shares that pay one dollar if an event happens and nothing otherwise. Someone has to say whether the event happened. On Polymarket, the largest such market, an *optimistic oracle* does: after a market closes, anyone may post an answer together with a bond, and the answer stands unless someone disputes it. A question disputed twice goes to a vote of the holders of UMA, the oracle's token. Holders who want to vote stake their tokens in the voting contract, each staked token counts once, and an answer backed by a threshold share of all staked tokens is final (Figure 1). Behind the vote, Polymarket keeps an administrator key that can settle a flagged question by hand.

The oracle's designers justify the vote with an economic argument: corrupting it should cost more than it pays. A coalition that wants to impose an answer would need a majority of the staked tokens, and the tokens would lose value once the manipulation showed. To *capture* a vote is to make it resolve to an answer of one's choosing. The argument grants that a large enough coalition could capture the vote and rests on what the capture would cost it. We check it against all 5,004 votes the oracle has held since it went live in 2023 (Section 4).

The prize is minted during the vote. A disputed question waits 72 hours at the median for its vote while its market keeps trading. Shares are minted in sets, a Yes share and a No share backed together by one dollar. A market's open interest is the number of sets outstanding, the most it can pay out. A buyer of the expected answer pays nearly a dollar, and the purchase mints a set whose other half, the answer a capture needs, sells for a tenth of a cent. We call a vote contested when the losing side held at least a tenth of the votes. Of the open interest in contested markets at the end of the vote's first round, 91% was minted after the vote was called, so a budget checked when a question is disputed misses most of what a capture would collect. When the question "Israel x Hezbollah ceasefire by April 18, 2026?" went to the vote, its market held $0.96 million; when the round ended it held $61.4 million, 7.5 times the value of the staked supply (Section 5.8).

The contracts pay a captured vote. We replay the capture of that vote on forks of Ethereum and Polygon. The round's 19 largest voters vote for the answer the market sold at a tenth of a cent, and the deployed voting contract, bridge, adapter and conditional-token contract settle the market on it, given one call by a permissioned resolver that can delay a payout but not change it. The 12 largest buyers of that answer during the vote are paid $18.4 million, 2.3 times the value of the staked supply. For those shares only the identity of the holders is counterfactual. The market's trades during the vote would have let the coalition itself hold 58.3 million such shares for $73,000 (Sections 5.8 and 7).

The largest prizes need the largest coalitions. The capture needs 63.6% of the stake because only 1.4% of it voted for that answer of its own accord. The same holds across markets: the other answer is cheapest when almost nobody believes it, so the largest prizes sit on the clearest questions, which only a coalition near the threshold can carry (Theorem 1 and Section 5.7). The designers' argument allows such a coalition and relies on the loss it would take, and the prize can exceed the value of the whole staked supply.

The limit. No counting rule changes this. Freeman, Lahaie and Pennock removed the jump from nothing to a dollar in markets settled by arbiters, paying each share the fraction of arbiters who vote for it [1], so that a coalition moves a payout only by its share of the votes. That does not make manipulation expensive, because no counting rule creates a cost; the cost comes from the token. We prove

that a vote protects the markets of a round against every coalition only while their open interest is below what a visible manipulation would cost the stake that votes. Paying clear outcomes in full lowers this limit by a factor that our rule attains exactly (Theorem 4 and Corollary 1).

On UMA the limit binds. The only cost the oracle itself imposes is a penalty of 0.1% of a voter's stake. With it even our rule protects markets of at most about $8,000, and only against coalitions too small to block a vote; every one of the 51 contested markets was larger. The ceasefire market lay beyond every counting rule's worst-case guarantee, even if manipulation had made the token worthless. Nor do past captures show the token adding much to the penalty: of 14 captures and attempted captures of on-chain votes, those that took nothing from the protocol left the governance token within its ordinary noise, and a capture of Polymarket's settlements takes nothing from UMA (Section 6.5).

Close votes. The margin of a vote is a second weakness. In March 2025 Polymarket asked whether Ukraine would agree to a minerals deal with the United States before April; the final agreement was signed on April 30 [2], [3]. The vote said Yes while the market priced Yes at about 17 cents. The rule then in force required half of all staked tokens, and Yes cleared that line by 3.81% of them after three wallets holding 22% of the stake revealed their votes in the last hour. Replayed on a fork of Ethereum, the vote rolls to another round without any one of the three (Section 4.4). Of the 51 contested votes that have decided a Polymarket payout since 2023, 12 would have rolled without the winner's largest supporter, though in only one case could that supporter have changed the answer alone (Section 4.3). UMA then raised the line to 65% of the staked tokens [4]. We prove that this moves the weakness: when the other voters are honest, the stake that can force an answer and the stake that can block one add up to a constant under every rule that settles a vote by counting it (Theorem 3). Quorum sizes in Byzantine agreement trade safety against liveness in the same way [5].

What an operator can do. Because the prize is minted after the question goes to the vote, the operator can stop it at the source. Halting a market's order book while its question is before the vote keeps the prize within the open interest of the dispute, which the operator can weigh against the stake before the vote; had trading stopped there, no standard-adapter market would have held more than the value of the staked supply (Section 5.8). Beyond that, the operator can cap what one round settles, add stake, or settle disputed questions by our rule, tally settlement: a dollar to an answer with four fifths of the votes and linearly less below that. Tally settlement would have paid the March Yes $0.64 a share and lets a coalition move a payout only in proportion to its stake, at the price of paying 11 of the 36 contested Yes-or-No winners less than a dollar (Section 6). A coalition that blocks a question until it is deleted hands its settlement to Polymarket's administrator (Section 7).

Contributions.

- The prize minted during the vote, measured for every contested market, and the capture of the largest replayed on forks of both chains, where the deployed contracts settle the market on the captured answer (Sections 5.8 and 7).
- A census of a deployed stake-voted oracle: the margin and the pivotal voters of all 5,004 votes, which of them decided a payout, the effect of the higher line, and both sides of the security budget for every contested market (Sections 4 and 5.7).
- The capture threshold, and an identity for every rule that settles a vote by counting it: when the other voters are honest, the stake that forces an answer and the stake that blocks one add up to a constant (Section 5).
- The limit of vote counting, attained by tally settlement among rules that pay clear outcomes in full, measured on UMA's contested markets with the cost of a visible manipulation estimated from 14 past captures and attempted captures of on-chain votes (Section 6).
- Two remedies replayed on the oracle's history, halting trading during the vote and tally settlement, a prototype against the live contract, and the reason a market-price fallback fails (Sections 6 and 8, Appendix B).

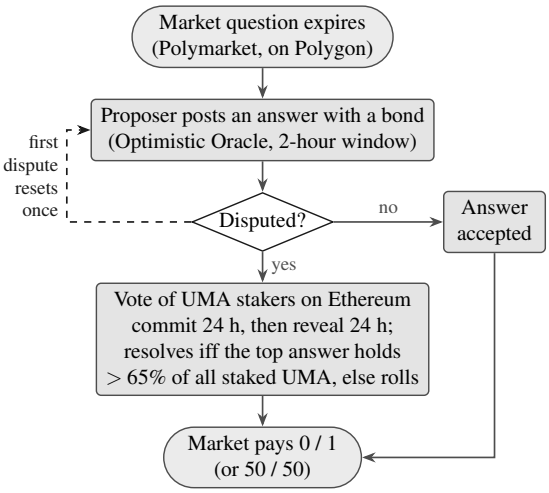


Figure 1. How a Polymarket question is settled. A proposer posts an answer with a bond to UMA's optimistic oracle on Polygon; an undisputed answer stands after two hours; a disputed one is reset once and, if disputed again, bridged to the Data Verification Mechanism (DVM) on Ethereum, where UMA stakers vote on the payout (the first dispute is voted on too, for its bonds only). The vote resolves only if the modal answer holds more than a threshold share of all staked tokens, otherwise it rolls to the next round and is deleted after four rolls. This paper measures what the vote settles, the margin by which it passes, and the limit of any rule that counts it.

2. Background

2.1. Settlement on Polymarket

Polymarket markets are Gnosis conditional tokens on Polygon: a binary market holds collateral in USDC and

TABLE 1. PARAMETERS OF THE DVM VOTE, FROM THE CONTRACT'S PARAMETER-CHANGE EVENTS.

Parameter	In force since	Value
agreement threshold θ	February 24, 2023	50% of staked
agreement threshold θ	April 20, 2025	65% of staked
participation floor	February 24, 2023	5,000,000 UMA
penalty λ, wrong or absent	February 24, 2023	0.1% of stake
rolls before deletion	February 24, 2023	4
unstaking delay	February 24, 2023	7 days

issues matching "Yes" and "No" shares, redeemable for one dollar each once a payout vector is reported [6]. The adapter that reports it [7] asks UMA's Optimistic Oracle [8]. A proposer posts a price, 1 for Yes, 0 for No, 0.5 for an unresolvable question, or a reserved "too early" value, with a bond; a two-hour liveness window follows; anyone may dispute by matching the bond. Every dispute is sent through the Polygon bridge to the DVM, but the adapter re-asks the question on its first dispute under a new request, so the vote on that dispute decides only who keeps its bonds. The market is paid on the new request: on its proposal if undisputed, on the DVM's answer if disputed again. A "too early" answer from the DVM makes the adapter re-ask later. The current adapter also has an administrator, who can pause a question and report any payout for it an hour after flagging it.

2.2. The DVM vote

The DVM is a commit-reveal vote of UMA stakers held in rounds of 48 hours, 24 to commit and 24 to reveal. Revealed votes are public as they arrive. Each staker's weight is her staked balance at the start of the round. A request resolves at the end of the round to the modal answer b^* if two conditions hold: the total stake that voted exceeds a fixed participation floor (5,000,000 UMA), and the modal answer's stake exceeds a fraction θ of *all* staked tokens, not of the votes cast. Otherwise the request rolls to the next round, and after four rolls it is deleted. On resolution every staker who voted for another answer or did not vote loses $\lambda = 0.1\%$ of her stake, and the pooled penalties are divided among the voters of b^* in proportion to stake. Nothing is paid or slashed in a round that rolls. The rule and its history are read from the contract source and from the parameter-change events it emitted (Table 1).

2.3. The vote of March 2025

Figure 2 shows the reveal day of the request that decided the minerals-deal market's payout, read from the `VoteRevealed` events. Panel (a) accumulates the revealed stake for the two leading answers over the day; the three circles are the three wallets. Panel (b) is the market's own price of a Yes share, by the minute, from six hours before the last reveal to one hour after the round ended. The two panels are the paper in miniature: the vote's margin was smaller than any of the three wallets, the market did not believe the answer until it was forced to, and the hour between the last reveal and the end of the round was a window in which anyone reading the chain could buy a share worth a dollar for less than 35 cents.

3. Threat model

Participants. A *staker* holds UMA in the voting contract and votes once per request; her weight is her stake. A *holder* owns shares of the market being settled. The same person can be both; the Wall Street Journal reported in May 2026 that most active UMA voters could be linked to Polymarket accounts [9], as summarized in [10]. The attacker we study is a *coalition*: it controls a share σ of the staked supply, may split it across any number of wallets, may abstain, may hold any position Q in the market being settled and in correlated markets, and sees every revealed vote as it arrives. It cannot change a commitment after the commit phase, cannot forge votes, and cannot read other voters' commitments. Polymarket's administrator can override the vote by flagging a question before its answer arrives, and a resolver can withhold settlement on the managed oracle (Sections 2 and 7); we study what the vote guarantees without either.

The rest of the electorate. We make no behavioral assumption about other voters. We describe them by two measured quantities: the *honest block* h_F, the share of the staked supply that votes the coalition's answer F for its own reasons, and the honest block h_T that votes the opposing answer T. Both are read off the tallies of past votes; they are not modeled. The remaining stake is called *rational* only in the sense that nothing is assumed about it; our cost-to-attack statements hold for every behavior of this remainder.

Goals and gains. In these terms capture makes the vote resolve to F, and a *stall* prevents any resolution. The coalition gains from capture through shares of F bought below a dollar, through the penalty pool that resolution transfers to the winning side, and through anything settled by the same answer elsewhere. It pays through the carrying cost of its stake: purchase and price impact for a newcomer, the token's loss of value after a visible capture for an incumbent, and the penalty λ if its side loses. Bribes paid outside the protocol and bugs in the contracts are out of scope; the vote's own rules are the object of study.

What is public. Stakes, votes, reveals, penalties and payouts are all on-chain. Our tallies use nothing else, and neither does the coalition.

4. A census of the vote

4.1. Data and method

Votes. We pulled every event of the voting contract (`VotingV2`, deployed February 24, 2023) from public archive endpoints up to October 6, 2026: request additions, rolls, deletions, resolutions, every revealed vote with its stake, every stake change, delegation and parameter change. Resolutions are tallied from their `VoteRevealed` events

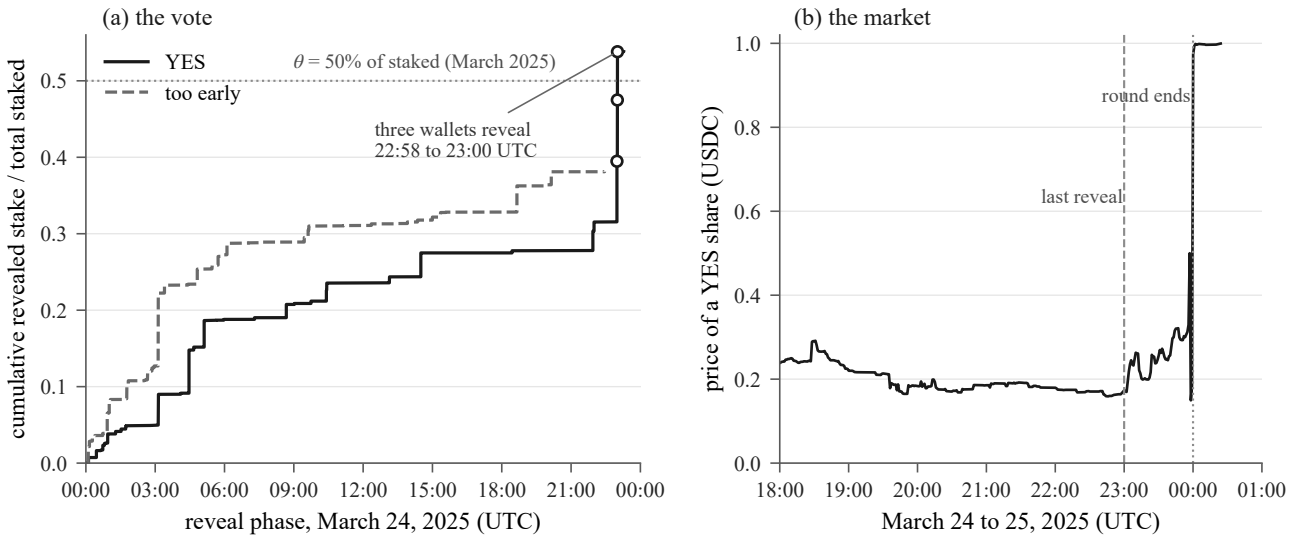


Figure 2. The March 2025 vote. (a) Cumulative revealed stake, as a share of all staked tokens, for "Yes" and "too early" during the 24-hour reveal phase of March 24, 2025 (UTC). The three circles are the three wallets that revealed between 22:58 and 23:00; the third took "Yes" past the 50% line, which it cleared by 869,647 UMA when the round ended. (b) The price of a "Yes" share on Polymarket by the minute. For nearly all of the hour after the last reveal the market sold Yes below 35 cents; the price reached 97 cents when the round ended.

in the resolving round, each request identified by its identifier, time and question text together, since requests that settle markets of one event share the first two; the staked supply of that round is read from the contract's frozen round record, which is the denominator the contract itself uses. For the two March 2025 requests we additionally decoded the penalty events the contract emitted and reproduced each voter's to the wei with a Python port of the contract's arithmetic (Appendix D).

Questions. Requests bridged after mid-2023 carry only "ancillaryDataHash:h,childBlockNumber:b". The question text, stamped by the oracle on Polygon, is in the event its bridge contract emitted at block b, and its Keccak-256 hash equals h; every contested request matched.

Payout votes. A Polymarket question's first DVM request comes from its first dispute, after which the adapter has re-asked the question, so that vote decides only the bonds; later requests of the question decide the payout. The rule agrees with the chain for all 162 contested requests matched to a market: only a first dispute's transaction re-asks the question. Of the 4,185 Polymarket questions that reached the DVM, 3,591 reached it only once, for the bonds of their first dispute. The other 194 votes are 26 on UMA's governance, 143 disputes of Across bridge bundles (a disputed bundle is deleted, so the vote moves only the bond), 17 questions asked on Ethereum and 8 assertions.

Markets. We match a question to its market without searching by title. Polymarket's adapter registers a question under $q = \mathrm{keccak}(a)$, where a is the text it sent to the oracle, and the conditional-token contract identifies the market as $c = \mathrm{keccak}(o \,\|\, q \,\|\, 2)$, with o the adapter's address and 2 the number of outcomes. The oracle appends ",ooRequester:o" to the text before forwarding it, so the text in the voting contract yields both a and o. We accept a derived c only if the conditional-token contract on Polygon reports two outcome slots for it. For 117 of the 121 contested requests matched this way, Polymarket's market API lists the market and returns the same q; it lists none of the other 4, whose markets exist on-chain. Neg-risk markets are prepared by a different contract under a different question identifier; we match those by exact title.

Open interest. A market's open interest at a block is the sum of its `PositionSplit` amounts minus the sum of its `PositionsMerge` amounts up to that block, from the conditional-token contract on Polygon; it is the number of complete Yes-and-No sets outstanding, each backed by one dollar. We take the last Polygon block whose timestamp is at most the end of the resolving round.

Prices. Token prices are Coinbase's public daily closes in UTC, so a close is the price at midnight UTC, when a voting round ends. Market prices are the market API's price history, by the hour for pre-dispute averages and by the minute around each vote's reveals. Every number in the paper is generated by one script from these sources.

4.2. Participation and concentration

The electorate is small and attentive. Across the 4,118 resolutions since the 65% change, the median request drew 518 voters holding 0.964 of the staked supply; absence is slashed, so almost everyone votes. The supply is concentrated. On October 6, 2026 the largest staker held 16.9% of 19,357,995 staked UMA, the four largest 37.12%, the ten largest 59.8%. No single address has ever held 35% of the supply (the maximum was 32.0%); the smallest group of top stakers whose combined holdings reach 35% has 4 members, and reaching 65% takes 13. The largest staker today is one of the three March 2025 wallets.

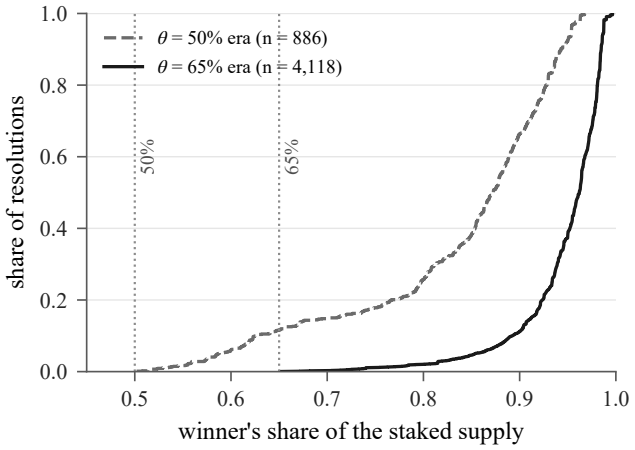


Figure 3. Winner's share of the staked supply at resolution, cumulative distribution over all resolved requests, before and after the threshold change of April 20, 2025. The dotted lines are the two thresholds.

4.3. Margins

Figure 3 shows, for every resolution, the share of the staked supply held by the winning answer, separately for the two eras. Most votes are near-unanimous: the median winner held 0.872 of the supply before the change and 0.962 after. The lower tails are where the oracle is decided. Before the change 103 of 886 resolutions had a winner below 65%, every one of which would roll under the current rule; the minimum was 0.502, the March vote. After the change the minimum is 0.651, 4 resolutions sit within two points of the line, and 46 within ten.

Of the 169 contested resolutions, 165 were on Polymarket questions and 51 of those decided a payout; the rest of this section counts those 51. The losing answers held a median 0.157 of the votes. The winner's slack above the requirement is 0.075 of the supply at the tenth percentile and 0.175 at the median, while the winner's largest single supporter holds 0.094 of the supply at the median and up to 0.292. Single-voter pivotality follows from these two distributions: in 12 of the votes the winner falls below the requirement without its largest supporter (6 before the change, 6 after; 57 of all 169 contested resolutions), and in all 28 since the change the removal of three supporters or fewer suffices. In 7 of the 12 the winner was Yes or No; in the other 5 it was "too early", and the largest supporter chose between two delays, asking the question again at once or voting on it again after a roll. Removing a supporter rolls a vote; carrying another answer is harder, and in only one of the 12 would the largest supporter's stake, voting for the runner-up, have made it win. The "too early" answer wins 15 of the 51, and in those the dispute was over whether the question could be settled yet.

4.4. The March 2025 vote in detail

The market was the subject of two requests in this round. The first came from the first dispute, after which the adapter had already re-asked the question; "Yes" won its vote by 0.19% of the stake with the same three wallets pivotal, and that vote decided only the dispute's bonds. The second request decided the payout, and Table 4 in Appendix D collects its reconstruction. The agreement requirement was $22{,}845{,}749/2$ UMA. "Yes" cleared it by 869,647 UMA. Without the three wallets, "Yes" would have held 31.6% of the supply and "too early" 38.1%; neither resolves, the request rolls. We checked this against the deployed code. On a fork of Ethereum mainnet taken one block before the first of the three reveals (block 22,119,898), we replayed every later reveal of the request with its original sender, calldata and timestamp: the contract resolves the request when the round ends, and it rolls the request if the reveal of any one of the three wallets is left out, even that of the smallest, which held 6.3% of the supply. Each wallet alone was pivotal. The three wallets' 5,083,491 UMA was 41.4% of the winning side, and the penalty pool of 10,553 UMA, drawn from the 8,706,371 UMA that voted "too early" and the 1,845,686 UMA that did not vote, paid them 4,364 UMA. At the day's price of $1.41 per UMA the staked supply was worth $32.2 million and the three wallets' stake $7.2 million.

What could capture have earned? A Polymarket market is a pool of complete Yes-and-No sets, each backed by one dollar, so no side can be paid more than the number of sets outstanding. When the round ended, 987,056 sets of this market were outstanding, read from the conditional-token contract's split and merge events on Polygon; the market's $7.08 million of volume was turnover. In the following week 960 wallets redeemed $1,129,633 of Yes shares, the five largest 44.6% of the total; the redemptions exceed the sets outstanding at the vote because new sets were created after it. Between the request reaching the vote and the last reveal, 260 wallets bought 486,583 Yes shares for $104,749. In the hour between the last reveal and the end of the round, when the outcome was readable on-chain, 65 wallets bought 177,792 for $72,566, at 16 to 34 cents for fifty minutes and up to 98 cents in the last ten.

Every vote has such an hour. Once the winner's revealed stake passes the line no other answer can reach it, so the outcome is fixed and public before the round ends: across the 36 contested Yes-or-No markets with minute prices, a median of 6 hours before. Usually the market had priced the answer already, above 90 cents in 35 of them when the deciding reveal landed. The exception is March 2025, where Yes stood at 17 cents when the outcome was fixed and the price caught up only as the round ended. A coalition need not hold a position before the vote: when the market is slow, it can buy one after its own reveal.

Who held the stake? The three wallets first staked on March 3, 2023, within 5.5 hours of each other and a week after the contract went live, and none of them staked tokens of its own again before the vote. The only additions were voting rewards, which two of the wallets re-staked 196 and 197 times, 188 times within two minutes of each other. The joint schedule, the common start and the reveal times are why we treat the three as one actor, and on that reading the capture was made with stake held for two years. Two

conclusions follow. A single market of this size cannot pay for a seven-million-dollar stake bought for the purpose, so the economics of capture are those of an incumbent, whose cost is the token's loss of value rather than a purchase (Section 5.7 measures that loss); and the mechanism's own penalty, however small, pointed the wrong way.

4.5. What the 65% change did

The change did what its proposal said it would: weak majorities now roll and give voters more time [4]. The proposal estimated that 6 of 291 requests in 2025 would have rolled under the new threshold; over the same year up to the change, our census counts 11 of 347 resolutions whose winner held 65% of the supply or less, consistent with that estimate. What the change did not do is make contested votes safer. Repeated rolls rose: before the change 25 requests rolled and 2 rolled twice; after it 40 rolled, 14 rolled two or more times, and one request rolled four times and was deleted (its question text was a hash the voters could not read, so no attack was involved, and it shows what a stall ends in: the question leaves the oracle). Single-voter pivotality continued, 28 cases, 6 of them on votes that decided a payout, with winners just above the new line and largest supporters of ten percent of the supply. Section 5 explains why: the threshold moves the capture requirement and the block requirement in opposite directions by the same amount.

5. Analysis

5.1. Model

A vote has a staked supply normalized to one. The coalition holds a share σ and votes F; an honest block h_F votes F; an honest block h_T votes T; the remaining share $m = 1-\sigma-h_F-h_T$ votes as it likes, and we write $y \in [0,1]$ for the fraction of it that votes F. The tallies are

$$x_F(y) = \sigma + h_F + y\,m, \qquad x_T(y) = h_T + (1-y)\,m.$$

Under the deployed rule with threshold $\theta \geq 1/2$, F resolves when $x_F > \theta$, T resolves when $x_T > \theta$, and the request rolls otherwise; at most one of the two can exceed θ. On resolution each losing or absent token pays λ, and the pool is shared by winning tokens in proportion to stake. These are the contract's rules; the only modeling choices are the normalization and the three blocks, and the blocks are measured from past tallies. We call the electorate *honest* when the remaining share abstains; its *participation* $\pi = h_F + h_T + \sigma$ is then the share of the staked supply that votes, and full participation is $\pi = 1$.

5.2. Three mechanical facts

Proposition 1 (Stall is free). A coalition holding at least $1-\theta$ of the staked supply that abstains prevents every answer from resolving; the request rolls each round and is deleted after four rolls; no penalty is ever charged to the coalition, because penalties are computed only for resolved requests.

Under the current parameters $1-\theta = 35\%$, and the four largest stakers hold 37.12%.

Proposition 2 (The penalty pays the winner). If b^* resolves with winning stake v^*, each winning token receives $\lambda\,(S-v^*)/v^*$, and a coalition holding σS inside the winning side receives the share $\sigma S/v^*$ of the pool $\lambda(S-v^*)$.

In March 2025 the pool was 10,553 UMA and the three wallets' share 4,364 UMA. The transfer is small; its direction is the point. The voters who resisted the capture financed it.

Proposition 3 (The narrow-coalition premium). A risk-neutral voter whose stake cannot change the outcome and who holds no position ranks the answers that can resolve by $p_b\,\mathbb{E}[\,S/v(b) \mid b \text{ resolves}\,]$, with p_b the probability that b resolves, not by p_b alone; the swing from a losing to a winning vote, pool reward plus penalty avoided, is up to $1/\theta$ times as large in a narrow win as in a landslide; and voting for such an answer strictly beats abstaining.

Pro-rata redistribution makes a win by a thin margin the most rewarding vote to join. Proofs of the three propositions are in the artifact; the first two are direct readings of the rule and the third is a one-line expectation.

5.3. The capture threshold

Theorem 1 (Capture). If $\sigma > \theta - h_F$, then F resolves in the current round whatever the remaining stake does. If $h_T \geq 1-\theta$, then F cannot resolve for any $\sigma \leq 1-h_T$. Between these two regions the outcome depends on the remaining stake: if the remaining voters hold no position and value only the vote's transfers, "everyone remaining votes F" is an equilibrium when $1-h_T > \theta$ and no remaining voter holds more than $\theta - h_T$, and "everyone remaining votes T" is one when $1-\sigma-h_F > \theta$ and none holds more than $\theta-\sigma-h_F$.

The guarantee is the first sentence. The honest support for the coalition's answer is a subsidy: every point of h_F lowers the stake capture requires by a point. In March 2025, treating every non-whale "Yes" vote as honest support, $h_F = 31.56\%$ cut the requirement from one half to $\theta - h_F = 18.44\%$ of the supply, and the three wallets held 22.25%. Their excess over the requirement is, by construction, the 3.81% by which the vote was won; what the theorem adds is the subsidy, a guaranteed win for less than a fifth of the supply. Reading the non-whale "Yes" votes as followers rather than honest supporters only lowers the requirement further. We do not claim that the remaining stake follows the whale below the guarantee: the natural selection argument for coordination games does not go through here, because the state of the world only moves which outcomes are feasible, and a numerical check of the noisy game, in the artifact, confirms a continuum of equilibria below the guarantee. The paper's cost-to-attack statements use the guarantee.

5.4. What a threshold can and cannot do

Theorem 2 (The threshold window). Under the threshold rule, against an honest electorate in which a block $h_F \le \theta$ votes the coalition's answer F and everyone else votes T, a coalition of stake share σ makes F resolve whenever $\sigma > \theta - h_F$, and keeps T from resolving, so that the question rolls, whenever $\sigma \ge 1 - \theta - h_F$. Both are impossible exactly when $\theta \in [\sigma + h_F,\ 1 - \sigma - h_F)$, an interval that is non-empty if and only if $\sigma + h_F < 1/2$.

The penalty, the participation floor and the rewards do not appear, and honest support for the coalition's answer helps it on both sides. Today's four largest stakers hold $\sigma = 37.12\%$: no threshold protects a question from them once the honest support for their answer reaches 12.88%, and at $\theta = 0.65$ they capture a March-like question (requirement 33.44%) outright. The single largest staker (16.9%) cannot capture at 65% unless the honest support for its answer exceeds 48.1%, but it blocks any question on which more than 18.1% of the supply votes against the majority answer. In the census, 14 of the 28 contested payout votes since the change, and 27 of all 480 payout votes, had a winner below 65% plus 16.9% of the supply: a single stake the size of today's largest, voting the other way, would have rolled each of them. The 65% change made capture harder and blocking easier by the same fifteen points.

5.5. Beyond thresholds

Could a cleverer counting rule do better? We consider every rule that maps the pair (x, y) of shares voting F and T (abstention being $1 - x - y$) to F, T or roll, that is *monotone* (F stays resolved when x rises or y falls), *neutral* (symmetric in the two answers) and *non-trivial* (a unanimous vote resolves). For such a rule let θ_f be the infimum of the x at which F resolves when everyone votes, and let D be the width of the band of full-participation splits in which nothing resolves. Against an honest electorate, a block h_F voting F and everyone outside the coalition otherwise voting T, let $\sigma_{\mathrm{cap}}(h_F)$ and $\sigma_{\mathrm{blk}}(h_F)$ be the infima of the coalitions that make F resolve and that keep T from resolving.

Theorem 3 (Tally-only rules). For every monotone, neutral, non-trivial tally-only rule and every $h_F \le 1 - \theta_f$:

(a) $\sigma_{\mathrm{cap}}(h_F) = \theta_f - h_F$;
(b) $\sigma_{\mathrm{blk}}(h_F) = 1 - \theta_f - h_F$; voting F blocks whenever any split of the coalition's stake does, and under a rule that counts abstention as opposition so does abstaining;
(c) $D = 2\theta_f - 1$, and $\sigma_{\mathrm{cap}}(h_F) + \sigma_{\mathrm{blk}}(h_F) = 1 - 2h_F$ for every rule in the class.

Part (c) is the identity promised in the introduction; for larger h_F the honest vote alone keeps T from resolving. Every point of capture resistance costs exactly one point of block resistance, whatever the rule; the only thing a designer chooses is θ_f, which fixes where the deadlock band sits. Quorum sizes trade safety against liveness in Byzantine agreement in the same way [5]. There a faulty node can vote for two values, so a larger quorum buys two units of safety for each unit of liveness it gives up; a staked token votes once, and the trade is one for one. In particular, counting the threshold against votes cast instead of staked tokens changes neither requirement: it only decides whether a blocker may hide in abstention or has to vote for the losing side, and through the honest voters who abstain it moves each requirement by a point or two (at 96% participation, 32.5% against 34.2% for blocking with $h_F = 0$). The proof (Appendix A) is three monotonicity arguments; the appendix also records the variant in which the rest of the electorate may vote the blocker's own answer, which is where the denominator does matter and which honest voters do not produce.

5.6. The security budget, written correctly

The designers' argument is "cost of corruption exceeds profit from corruption". Theorem 1 and Proposition 2 give the terms. Let $C(\sigma)$ be the cost of carrying a share σ of the supply through one vote (two days of voting and seven days of unstaking): price impact plus the token's subsequent loss of value for a buyer, the loss of value alone for an incumbent. Let $\bar{p}$ be the coalition's average purchase price of claims paying a dollar on F, here or elsewhere, and $Q_{\max}$ the number it can buy at that price; its gains are these claims and the pool. In the limit $\sigma \downarrow \theta - h_F$ of the coalitions that capture for sure, which exist when $h_F \le \theta$ and $h_T < 1 - \theta$, cost exceeds the largest gain if and only if

$$C(\theta - h_F) \;>\; (1 - \bar{p})\, Q_{\max} \;+\; \frac{(\theta - h_F)(1 - \theta)}{\theta}\, \lambda V, \quad (1)$$

with C continuous and V the market value of the staked supply. Condition (1) is thus necessary for safety against guaranteed capture, up to equality, and it is sufficient when $C'(\sigma) \ge \lambda V(1 - \theta)/\theta$ for $\sigma \ge \theta - h_F$, as when every staked token loses a common fraction of at least $\lambda(1 - \theta)/\theta$ of its value (Appendix A). Three things differ from the folk version. The cost side carries $\theta - h_F$, not one half; the profit side grows with how obvious the lie is, since $1 - \bar{p}$ is largest when the coalition's answer is cheapest; and the penalty appears on the profit side. In March 2025 the right-hand side was bounded by the $0.99 million of open interest at the end of the round plus 4,364 UMA of penalties, and the left-hand side was whatever a visible capture cost the holders of $7.2 million of stake in the token's value.

5.7. The budget, measured

We read both sides of (1) from the chain for every contested Polymarket question. The profit side is bounded by the market's open interest when the vote closed: no side can be paid more than the number of complete sets outstanding. Of the 165 contested Polymarket votes, 162 match their market (Section 4.1), 121 through the market identifier derived from the question text and the rest, neg-risk markets, by exact title. Of these, 51 decided a payout

and form 51 distinct pairs of market and round; the other 111 were first disputes. The cost side is the stake that would have carried F, the side the vote did not choose (the side with more votes when the vote said "too early"). By Theorem 1 an incumbent needs $\theta - h_F$ of the supply, with h_F the share F actually received, and we value that stake at the token's closing price on the day the vote closed. The ratio r of open interest to that value bounds the fraction of the capture stake's value that a capture can destroy and still pay.

Figure 4(a) shows both sides. The median market held \$1,160,000 of open interest, the median capture stake was worth \$13.0 million, and the median r was 8.7%. The tail is not small. In 24 market-rounds the open interest exceeded a tenth of the capture stake's value, and in 5 it exceeded all of it: there, even the token's total loss would not have ruled out a profitable capture.

The tail has grown. Before the 65% change 8 of 23 markets had $r \geq 0.1$; since, 16 of 28. The higher threshold raised the stake a capture needs by fifteen points, but over the same period the token fell from \$1.41 at the March vote to \$0.45 at the end of the census, taking the value of the staked supply from \$32.2 million to \$8.6 million, while the largest markets grew. When the vote on "Israel x Hezbollah ceasefire by April 18, 2026?" closed, its market held \$61.4 million of open interest, 7.5 times the value of the staked supply; the stake its capture needed was worth \$5.2 million, and the staked supply \$8.1 million. Table 2 lists the ten largest ratios.

Who could have collected? The 5 markets with $r \geq 1$ each needed at least 49.9% of the supply, because almost nobody had voted the other side; the ten largest stakers today hold 59.8% together. A coalition the size of today's four largest stakers (37.12%) could have carried the other side in 6 market-rounds, with ratios of at most 0.52, and the largest single stake (16.9%) in none. The budget fails where the designers' argument looks, at coalitions of half the stake or more valued at market prices, and not for today's largest stakers acting alone. What a single stake decides is whether a vote resolves (Section 4.3).

Open interest overstates what a coalition can collect, since it must buy F from holders willing to sell, so we also measure what it could have bought. A coalition that means to carry F buys it while it is cheap and before its own reveal. Taking each market's actual trades from the request reaching the vote to the decisive reveal, we let the coalition take the place of the wallets that bought F on net, at their own average prices, so that a share bought and resold inside the window counts once. If F is then paid a dollar, the coalition collects \$523,000 in the median market, 53% of the open interest, and up to \$58.2 million; this executable revenue exceeds a tenth of the capture stake's value in 18 market-rounds and all of it in 2.

An incumbent should compare it with the whole stake it puts at risk: a coalition the size of today's four largest stakers could have collected at most 2.8% of its stake's value from any market it could have captured, and one the size of today's largest could have captured none. Open interest says where no loss of stake value could guarantee safety; the executable revenue says where a capture would have paid at the trades that took place; whether any capture did pay would need the coalition's identity and positions, which the chain does not reveal. Capturing a single contested market pays such a coalition only if a visible capture costs its token less than a few percent. A buyer of stake, conversely, pays more than the market value of the capture stake, since buying moves the price.

The cost side for an incumbent is the token's loss of value, and the March vote shows what that loss was. The three March 2025 wallets held stake worth \$7.2 million. The market's trades would have let them collect \$264,100 on Yes: \$73,700 before the reveals began, \$122,700 before their own reveals, and \$67,700 after them, when the outcome was fixed on-chain and the market still sold Yes cheaply. If the three were one actor, the capture paid if it cost the token less than 3.7% of its value, less than one day's standard deviation of UMA's return relative to ETH (4.4 points), and the token showed no loss of that size (Appendix E): its price could not have told anyone whether the capture paid. An oracle whose safety rests on its incumbents not wanting to capture is not secured by its slashing parameters, and the market for its token does not price the difference.

5.8. When the prize is minted

Most of the open interest measured when a vote closes did not exist when the vote was called. A disputed question waits 72 hours at the median for its round to end while its market trades on: buyers of the expected answer near a dollar earn a fraction of a percent, and each purchase mints a set whose other half sells near the minimum price of a tenth of a cent. A budget checked when the question is disputed therefore weighs the stake against a prize that is still growing. For the 43 standard-adapter market-rounds we read the open interest at the block of the dispute that sent the payout vote's request to the DVM and at the end of its first round, on which tally settlement pays. It grew from \$15.3 million to \$177 million, so 91% was minted after the vote was called; the twenty largest markets grew 12-fold at the median, against 3.5-fold over an equal span before the dispute. No market had $r \geq 1$ when its vote was called, and 3 did when the round ended. Halting a market's order book while its question is before the DVM stops this minting (Section 6.6): with the open interest of the dispute block, the loss that tally settlement needs (Figure 4(b)) would have stayed below the value of the staked supply for every market and below 14% for all but 2, though a coalition could still buy before disputing.

The ceasefire market shows it at full size: called with \$0.96 million of open interest, it held \$25.0 million a day later and \$61.4 million when the round ended, 64 times as much. Before the reveals, buyers paid 99.87 cents on average for 55.5 million "Yes" shares, and 1,285 wallets bought 52.9 million "No" shares on net, 85% of the volume at a tenth of a cent and the largest buyer about 10% of it. A coalition in the place of these buyers, at their prices, would have held 58.3 million shares for \$73,000 up to the

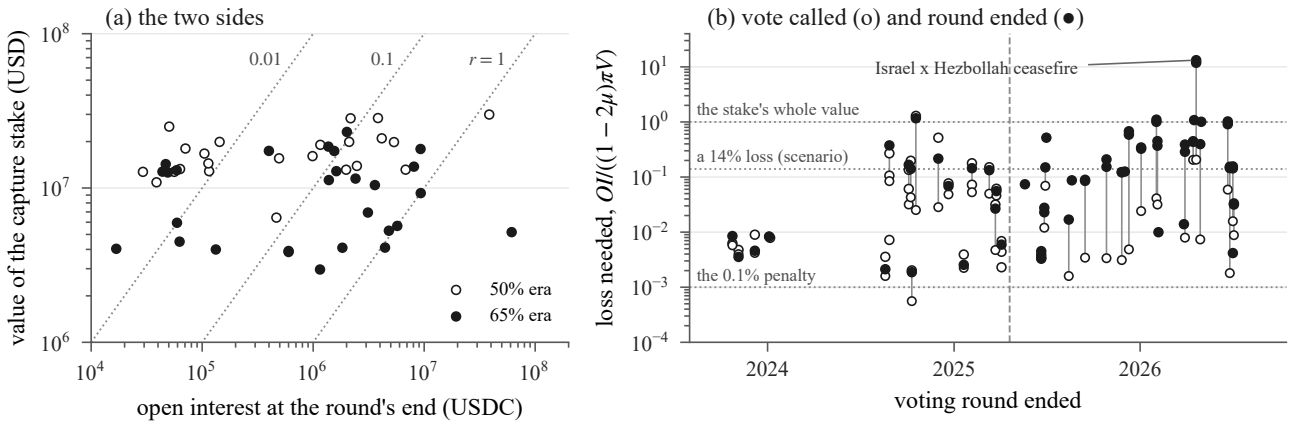


Figure 4. The security budget of the contested votes that decided a payout. (a) Open interest when the vote closed against the value of the capture stake, the share $\theta - h_F$ that would have carried the other side; dotted lines mark r of 1, 0.1 and 0.01. (b) Per round, the loss $OI/((1-2\mu)\pi V)$ that tally settlement with $\mu = 0.2$ needs at the participation π of the round's first vote (Corollary 1), with the open interest when each standard-adapter market's vote was called (hollow) and when the round ended (filled); the dashed line is the threshold change.

TABLE 2. THE TEN CONTESTED PAYOUT VOTES WITH THE LARGEST RATIO r OF OPEN INTEREST WHEN THE VOTE CLOSED TO THE MARKET VALUE OF THE CAPTURE STAKE; AMOUNTS IN MILLIONS OF DOLLARS.

Question	Closed	θ	Vote	F	h_F	OI	Stake	r
Israel x Hezbollah ceasefire by April 18, 2026?	2026-04	65%	Yes	No	1.4%	61.4	5.2	11.86
Will Trump launch a coin before the election?	2024-10	50%	Yes	No	0.0%	38.6	29.9	1.29
Will "Donk" be said during the PGL Bucharest 2026 Grand Final?	2026-04	65%	No	Yes	14.5%	4.4	4.1	1.08
Clavicular pregnancy in 2026?	2026-04	65%	Yes	No	0.0%	5.7	5.7	1.01
Tesla launches unsupervised full self driving (FSD) by June 30?	2026-02	65%	Yes	No	0.2%	9.3	9.3	1.01
Will Trump agree to withdraw troops from the Iranian region by June 3…	2026-06	65%	Yes	No	0.0%	4.8	5.3	0.91
Trump declassifies UFO files in 2025?	2025-12	65%	Yes	No	0.0%	8.1	13.8	0.59
Fordow nuclear facility destroyed before July?	2025-06	65%	Yes	No	0.5%	9.3	17.9	0.52
Israel x Hezbollah Ceasefire in 2024?	2024-11	50%	too early	Yes	28.3%	6.8	13.1	0.52
Will Trump talk to Vladimir Putin in January?	2026-02	65%	No	Yes	16.9%	3.1	6.9	0.45

decisive reveal (the market charged no fees), a claim to $58.3 million settled by its own votes. With 1.4% of the stake voting "No" honestly, it needed 63.6% (Theorem 1), held by the round's 19 largest voters, the largest with 8.6%. Charged the value of the staked supply, $8.1 million, the capture pays up to 7.2 times its cost net of the purchase; half the shares at ten times the price would still pay 3.5 times, and the capture loses only for a coalition holding more than 0.96 of all UMA in existence, staked or not.

5.9. What the roadmap does not fix

UMA's published plan for the next oracle generation names three directions: dynamic bonds that scale with a market's size, resolution by several tokens, and detection of bribery through an external restaking layer [11]. Bonds live at the proposal stage and do not enter the vote; a second token changes S and not $\theta_f - h_F$; and bribery detection presumes a bribe to detect, whereas the market being settled is itself the bribe contract, paid in shares to whoever votes the right way, indistinguishable from trading. All three stay inside the class of Theorem 3.

6. The limit of vote counting

Theorem 3 closes one door. A rule that either resolves or rolls draws a line, and wherever the line is drawn, the stake that captures and the stake that blocks add up to the same constant. Freeman, Lahaie and Pennock removed the same jump in arbitrated markets by paying each share the fraction of arbiters who vote for it [1]. A rule can bound how far a coalition moves a payout, but it cannot make moving it costly; that cost is the stake's loss of value, and no rule protects a market larger than the loss allows (Corollary 1). Tally settlement is the rule that reaches the limit while paying clear outcomes in full.

6.1. The rule

Definition 1 (Tally settlement R_μ). Fix $\mu \in [0, 1/2)$. A disputed question is settled on the tally of its first DVM round, whether or not the DVM resolves the request in that round. If "too early" alone leads the tally, or no vote is cast, the question is asked again, as today. Otherwise let f be the share of the votes cast for Yes,

a vote for one half counting half; a Yes share is paid

$$\rho(f) = \min\Big\{1, \max\Big\{0, \frac{f-\mu}{1-2\mu}\Big\}\Big\}$$

and a No share $1-\rho(f)$. An undisputed question pays as the oracle says.

With $\mu = 0$ this is the proportional rule of [1]. A vote for "too early" counts against Yes, since a Polymarket question resolves No unless Yes is established by its deadline: a minority saying that it is too early to settle lowers what Yes is paid, and a plurality has the question asked again. A vote won with $1-\mu$ of the votes cast pays exactly as today; below that, the payout follows the tally. Polymarket already pays some questions at one half, and the conditional-token contract already accepts fractional payout vectors; tally settlement makes that outcome continuous in the tally. With $\mu = 0.2$, the "Yes" of March 2025, which held 58.5% of the votes cast, would have been paid 0.64 a share (Figure 5 in Appendix C). Without the three wallets "too early" would have led the round with 55% of the votes, and the question would have been asked again. The rule reads only the votes. Appendix B explains why the obvious alternative, paying a contested vote at the market's own pre-dispute price, is not safe: a coalition that can force such a fallback buys the reference price back at no cost at settlement (Proposition 5).

6.2. What the rule guarantees, and the limit

Theorem 4 (Tally settlement). Against an honest electorate with participation $\pi > 0$ that votes Yes or No, a block h_F for the coalition's answer F and a block h_T for the other:

(i) whatever the coalition does short of having the question asked again, F is paid between $\rho(h_F/\pi)$ and $\rho((h_F+\sigma)/\pi)$, and every amount in between is within its reach; so a coalition moves any share's payout by at most $\sigma/(\pi(1-2\mu))$, and a roll gains it nothing;

(ii) the coalition can have F paid one dollar if and only if $h_F + \sigma \geq (1-\mu)\pi$, at full participation $\sigma \geq 1-\mu-h_F$;

(iii) among monotone, neutral rules that pay a dollar to any answer with at least $1-\mu$ of the votes cast, none has a smaller worst-case shift per unit of stake at participation π than $1/(\pi(1-2\mu))$, which ρ attains; a rule whose payout jumps, as every rule that pays only zero or one does, is moved by the whole jump with an arbitrarily small stake;

(iv) a coalition holding Q_j shares of its answer in each market j that a round settles gains at most $\sigma\sum_j Q_j/(\pi(1-2\mu))$ from its votes, with π the smallest participation among those markets; if a visible manipulation costs a stake the fraction δ of its value after penalty-pool rewards, it does not profit when the open interest of those markets totals at most $\delta(1-2\mu)\pi V$, with V the market value of the staked supply;

(v) by voting "too early" the coalition can have the question asked again if and only if $\sigma > \max\{h_F, h_T\}$, at full participation $\sigma > \max\{h_F, (1-h_F)/2\}$.

The second half of (iii) is the observation of [1] for binary outcomes, in the language of stake: a line is a point of unbounded leverage, and the census found the oracle's contested votes sitting next to one. The first half prices full payment for clear outcomes, since the proportional rule moves a payout by $1/\pi$ per unit of stake and the ramp by $1/(\pi(1-2\mu))$. Part (iv) is what the ramp does to the budget of Section 5.6: the condition no longer involves the margin, the honest support or the size of the coalition, only the open interest a round settles against the value of the stake that votes, and it is tight for a coalition that holds the whole other side. Part (v) is the delay the rule keeps: at full participation, $\theta = 0.65$ and $h_F = 0.1$ it takes 45% of the stake, where 25% rolls the deployed vote (Theorem 2). The proof is in Appendix A; parts (i), (ii) and (v) were also checked by exact enumeration of every coalition split and every honest abstention on a grid.

Parts (iii) and (iv) together give the limit of vote counting. We say that a rule protects the markets of a round at loss δ and participation π if no coalition gains from manipulating them, whatever the honest electorate with that participation and whatever part of their open interest it holds; protection is a worst-case guarantee.

Corollary 1 (The limit). Suppose a visible manipulation costs a stake the fraction δ of its value after penalty-pool rewards, let V be the market value of the staked supply and $\pi > 0$ the share of it that votes. For every monotone, neutral rule that pays a dollar to any answer with at least $1-\mu$ of the votes cast, some honest electorate and some coalition gain from manipulating a market whose open interest exceeds $\delta(1-2\mu)\pi V$. Under tally settlement, no coalition gains from manipulating the markets that one round settles when their open interest totals at most $\delta(1-2\mu)\pi V$, with π the smallest participation among them.

A vote therefore protects the markets of a round only up to what a visible manipulation would cost the stake that votes, πV, and paying supermajorities of $1-\mu$ in full multiplies that limit by $1-2\mu$. With $\mu = 0$, proportional settlement protects up to $\delta\pi V$, and no counting rule protects more. The proof is in Appendix A.

Voters with positions, and disputes. The coalition of Theorem 4 is any set of voters, so voters who hold positions move a payout together by at most their combined stake divided by $\pi(1-2\mu)$; under the deployed rule they move nothing unless one of them is pivotal, and then a dollar. A dispute changes what a share pays only if the side it favors ends with more than μ of the votes cast, so disputing a clear outcome gains nothing, and a disputer with stake gains at most the bound of part (i).

6.3. What the rule would have paid

Because the rule needs only tallies, we replay it over every Polymarket vote that decided a payout, on the first round of each (Table 3 in Appendix C). Of the 457 Yes-or-No resolutions both rules pay, tally settlement with $\mu = 0.2$ pays 15 less than in full (3.3%), by a median of 20.5 cents, 10 of them since the 65% change; proportional settlement pays 332 less than in full, mostly by a fraction of a cent. Of the 36 contested Yes-or-No resolutions, the winners would have been paid a full dollar in 25 and 0.74 or more in nine of ten. At full participation a tenth of the supply moves any payout by at most 16.7 cents, and a one-dollar capture of a March-like question needs 48.44% of the supply. In the 7 single-voter-pivotal Yes-or-No resolutions the largest supporter moves the winner's payout by a median of 3.9 cents and at most 6.7, since the roll it could cause no longer moves the payout.

Coalitions the size of today's largest stakes show the trade the rule makes. In the 19 contested Yes-or-No payout votes since the change, a stake the size of today's largest (16.9%) that switches from the winner to the other side rolls 9 of them under the deployed rule and flips none; under tally settlement it moves the winner's payout by a median of 21 cents and at most 30. A coalition the size of the four largest rolls all 19 and moves the payout by a median of 57 cents and at most 66. The deployed rule lets such stakes delay; tally settlement lets them move money, in proportion to their size and never by the whole dollar. They can still delay by voting "too early" (Theorem 4(v)): moved to that answer, the largest stake would have had 3 of the 19 questions asked again, and the four largest 12. The rule's price is the haircut itself: a vote that is right but contested pays its winners less than a dollar, by an amount set by the share the losers held.

6.4. The limit on UMA's markets

Corollary 1 turns the budget of Section 5.7 into one number per market: the loss of stake value δ a market needs before any rule that pays supermajorities in full can protect it at the participation of its first round. In the 51 contested payout votes it ranged from 61.5% to 98.6% of the staked supply, with a median of 93.7%. The only loss the oracle imposes is its penalty and the pool reward a manipulator forgoes, at most λ/θ of a voter's stake in all, and a coalition large enough to block avoids both (Remark 2, Appendix A). With $\delta = \lambda/\theta$ the limit is at most about \$8,000 at the end of the census, and all 51 contested market-rounds of Section 5.7 held more. Beyond the penalty, δ is the token market's verdict on a manipulation, which Section 6.5 measures. A manipulation changes the coalition's take from the penalty pool by at most λ/θ of its stake per request (0.15% at $\theta = 0.65$), which moves each of the two counts at a tenth below by at most one market, the count at the scenario of Section 6.5 by at most two, and no other count. With $\mu = 0.2$, the median market is safe once δ exceeds 8.5%; 25 markets need more than a tenth, 13 more than a quarter, and 5 more than the value of the staked supply. The ceasefire market of Section 5.7 is beyond the worst-case guarantee of every counting rule, proportional settlement included, even with $\delta = 1$. Counted by round, as the limit requires, those needing more than a tenth rise from 25 to 29, and those beyond the value of the staked supply remain 5.

Fee-funded payments for truthful votes [1] would have needed a pool of at least \$831,000 to deter every coalition on the March 2025 market under the coalition loss that Remark 1 stipulates, against the \$15,000 the penalty pool paid.

6.5. What a visible manipulation costs

To measure δ we take the captures and attempted captures of on-chain votes in the catalogue of Feichtinger et al. [12] that have a public date and a year of daily prices, and two later ones, Compound's treasury vote of July 2024 [13] and UMA's of March 2025; Appendix E gives the method and the 14 cases. Where a capture drained or diluted the protocol itself, the governance token lost a median 25% of its value against ETH in the week, significantly in 3 of 6. A capture of Polymarket's settlements takes nothing from UMA, and none of the 8 captures and attempts that took nothing from their protocol had a significantly negative week; their median loss was 4.2%, and the one-sided 95% bound on their mean is 13%. The sample detects a mean loss of 14% only four times in five, so 14% serves below as a scenario for UMA. At that loss 17 of the 51 contested markets lie beyond the rule's guarantee, and in 13 of them the shares a coalition could have bought from the market's own trades exceed the limit by themselves.

6.6. What follows for deployment

The first remedy works on the prize. Polymarket's operator matches orders off-chain and settles the matches on-chain [14], so halting a market's order book once its question goes to the DVM stops the purchases that mint new sets. Anyone can still split collateral into sets on the conditional-token contract, but a set returns exactly its cost under any payout, so it gains a coalition nothing unless the other half can be sold. The prize is then at most the open interest of the dispute block, which the operator can weigh against the stake before the vote (Section 5.8), at the price of 72 hours at the median without trading in a disputed market. A coalition's stake moves every question in a round, and anyone can bring a question to the vote by disputing its proposals (Section 2), so the open interest to weigh is the total that one round settles: choose the loss of stake value δ to rely on and cap that total at $\delta(1-2\mu)\pi V$, or add stake; no counting rule protects more. Asking a question again when less than a share π_0 of the staked supply votes in its first round makes the cap hold with π_0; the DVM's own participation floor does not, since tally settlement pays whether or not the DVM resolves. Settle disputed questions by the tally, with μ read off Table 3: among rules that pay clear outcomes in full it reaches the limit, and the threshold change of April 2025 then stops mattering for what a share pays. The rule needs one new input, the first round's tally by answer, and no

market data. It applies as built to the standard adapter; Polymarket's neg-risk operator accepts only the payouts $[1, 0]$ and $[0, 1]$, so those markets need a contract change first.

7. Replays on the deployed contracts

The capture. On a fork of Ethereum at the end of round 10281's commit phase, the 19 largest voters on the ceasefire question recommit "No" with their own salts, and the round's 496 reveal transactions for the question are replayed with their senders, blocks and timestamps, the coalition's with only this question's vote changed. The live voting contract resolves "No", and the live root tunnel publishes it. This market is on the managed oracle, as are 14 of the 51 contested payout votes, all since October 2025: the adapter cannot resolve until a resolver role settles the request, a call that passes no price. On a fork of Polygon before the historical delivery, the simulated steps are the validators' system call that hands this message to the FxChild contract and the settlement by the resolver that settled this request on chain; the deployed child tunnel, oracle, adapter and conditional-token contract pay "No", and the vote's 12 largest "No" buyers redeem 18.4 million shares for $18.4 million. Unchanged, the same replay pays them nothing; under capture the coalition's stake rises by 21,769 UMA over the round. The $18.4 million is 2.3 times the value of the staked supply. The replay checks the two facts that the counterfactual of Section 5.8 assumes: by the contract's own accounting the coalition's stake carries the round, and no contract check between the vote and the payout stops a captured answer. Two roles remain: the resolver, who can delay the payout by withholding its call but cannot change the answer, and the adapter's administrator, who can flag the question before the answer is delivered and settle it by hand an hour later (Section 2); the replay leaves the administrator out.

The block. The latest contested payout vote whose answer pays, "Will Donald Trump publicly insult someone on June 21, 2026?" ($601 thousand of open interest), is forked the same way. Its four largest voters, 35.8% of the stake, vote "Yes"; "No" falls to 54.5% and the question rolls. In constructed later rounds every voter votes as before; the contract rolls the question four times and deletes it in the fifth round, charging no one for it. On Polygon the adapter then cannot resolve, and the collateral stays locked until an administrator of the adapter flags the question and, an hour later, reports a payout of its choosing. A coalition that can block hands the settlement to the operator's key.

8. Prototype

Tally settlement needs the vote's tally on Polygon, where the adapter runs; the tallies live in the voting contract on Ethereum. The prototype (Solidity 0.8.24, Foundry [15]) has three parts. *MarginReader* (Ethereum) takes a resolved request's identifier and index, checks one against the other, and stores the winner's share in basis points for the bridge. Read against the staked supply it runs on today's contract: on a mainnet fork it returns 5018 and 5380 basis points for the two March 2025 requests and rejects a wrong index. The version the rule needs reads Yes's share of the votes cast, a vote for one half counting half, or a flag to ask again when "too early" led the round; it needs one view, the round's tally by answer, which the deployed contract stores but does not expose, and on a mock with the March tallies it returns 5853 basis points. *TallyAdapter* (Polygon) pays a question settled without a vote as the oracle says and a disputed one $\rho(f)$ to Yes and the rest to No; ten tests cover the March tallies (6,421 basis points to Yes), the full-payment threshold, the one-half and ask-again cases, and a fuzz test of the bound $\sigma/(1-2\mu)$. Two tunnels carry the share to Polygon (Appendix G). A disputed question costs about 102,204 gas on Ethereum once and at most 155,719 on Polygon once, and nothing recurs per trade (Appendix G).

9. Related work

Rational security of deployed protocols. Our method follows the work that models a deployed system's participants as rational, finds the equilibrium the designers did not intend, and repairs the mechanism: selfish mining under fee-driven rewards [16], [17], MEV-aware consensus [18], symmetric and personal random reporting rewards [19], bribery of proof-of-stake validators and the counterincentives of slashing [20], order-policy enforcement [21], payment channels [22] and fees under parallel execution [23]. Relative to the closest, the bribery analysis of [20], the market being settled supplies the bribe, so the profit side of the budget is observable and the honest block sets the cost side. Budish shows that the cost of securing a blockchain must grow in proportion to what an attack would gain [24], and Budish, Lewis-Pye and Roughgarden ask when consensus can be expensive to attack in the absence of a collapse of the token's value [25]; Corollary 1 is the counterpart for a vote that settles markets, where the deterrent beside a 0.1% penalty is the token's loss.

Schelling-point oracles and bribery. Contingent bribes of Schelling-point votes are Buterin's $p+\epsilon$ attack [26], Dal Bó's pivotal bribe of a committee [27] in token form, and UMA's roadmap names them as the open problem [11]; we add that no briber is needed when voters can hold the asset being settled and that the required stake is the margin $\theta - h_F$. Freeman, Lahaie and Pennock pay each share of an arbitrated market the fraction of arbiters voting for it, observe that any binary settlement has a point where one arbiter moves a share by a dollar, and fund peer-prediction payments from trading fees [1]. Tally settlement is their proportional rule with full payment for clear outcomes; Theorem 4 adds the cost of that constraint, the bound for stake-weighted coalitions and the cap, and Remark 1 prices their payments on a contested Polymarket market. Self-resolving markets pay their own final price [28], and Appendix B shows why that idea fails for contested votes. Deng, Tao and Wang show that past a threshold share of antagonistic voters no anonymous mechanism implements

the informed majority's decision [29]; Theorem 3 is the corresponding statement for deterministic stake-weighted tallies against an honest electorate, with the exact trade between the stake that captures and the stake that blocks, which quorum sizes govern in Byzantine agreement [5].

Oracle and market manipulation. Price oracles in decentralized finance are manipulated by trading against the markets they read [30], [31]; here the oracle is a vote and the market is what it settles. Proposition 5 is the manipulation of a cash-settled contract through its settlement index [32], with a vote in place of the index. A manipulator who moves a prediction market's price can improve its accuracy [33]; a capturing coalition changes the payout and leaves the price alone. Section 6.5 draws on the catalogue of attacks on DAO governance of [12]; Kitzler et al. find a significant loss after 12 of 22 crimes against DAOs, 13.5% on average across those 12 [34].

Other resolution designs. Kleros decides by plurality of a stake-drawn jury and escalates through appeals [35], [36], Augur by escalating REP bonds that end in a fork of the token [37], and Reality.eth by doubling bonds with a designated arbitrator [38]; a single Kleros round is a tally-only rule, while appeals and bonds leave the class of Theorem 3 (Appendix F). UMA's votes have been measured by journalism, which reports that nine addresses dominate disputes and that most active voters hold Polymarket accounts [9], [10], [39], [40], [41], and by studies of whether language models reproduce its resolutions [42] and whether disputed markets price the event or its adjudication [43]; voting power has been measured for DAOs [44]. The census adds the margin of every vote, which votes decide a payout, the pivotal voters, and both sides of the security budget.

10. Discussion

Limitations. The three March 2025 wallets are treated as one actor on the strength of their timing and staking history, without funding-chain evidence. Honest blocks are read off votes, which conflates honest belief with any other reason to vote a fixed way; the capture guarantee does not depend on the distinction, the realized requirement does, and voters who follow the coalition only lower it. Executable revenue uses the trades that took place, with the coalition in the place of the net buyers at their prices; buying more would have moved the price. Conversions may inflate the open interest of the 8 neg-risk market-rounds; without them 16 of the 17 markets beyond the limit at 14% remain, and all 5 beyond $\delta = 1$. The tally-settlement replay keeps the votes outside the replayed coalitions as cast; voters who know that the payout follows the tally may vote, abstain or dispute differently; Corollary 1 covers abstention through π and counts voters with positions who change their votes as part of the coalition. The view exposing the votes cast is not built, and the tunnels are tested against a mock bridge. The fork replays run one round of real votes; the later rounds of the blocking replay are constructed, and the bridge's delivery and the resolver's call are simulated. With more than two answers, honest votes for any answer other than the majority's count toward h_F when blocking.

Ethics. Every input to this paper is public: on-chain events, public APIs and published documentation. Addresses are pseudonymous and we do not attempt to identify their owners; the attribution of three wallets to one actor is stated as an inference. No transaction was sent to any live contract; the counterfactuals are computations, and the replays and the prototype ran on local forks and in unit tests. The weaknesses discussed follow from parameters the operator published and from a vote the operator and the press have already analyzed; the paper adds measurement and a design analysis. Before submission we reported the measurements, the analysis and the remedies to Polymarket through the private vulnerability-reporting channel of its adapter repository on October 9, 2026 and, since the remedies on the oracle's side are governance decisions, posted them on UMA's governance forum, where its parameters are proposed, on October 9, 2026; we offered both the artifact.

AI disclosure. Generative AI tools, including Codex and Claude, assisted with literature searches, coding assistance and prose polishing. The authors conceived the paper's core ideas, developed its methods, verified the correctness and originality of all content, including the references, and take full responsibility for it.

Artifact. The anonymized artifact contains every script with the decoded per-request summaries, the exact-arithmetic checks, the slashing port, the replays on forks of both chains, both prototypes with their tests, and the proofs.

11. Conclusion

A stake-voted oracle is safe only while the markets it settles are worth less than what a manipulation would cost its stake, and on the oracle that settles the largest prediction market those markets grow after their questions go to the vote. In April 2026 one market grew 64-fold during its vote to 7.5 times the value of the staked supply, and on forks of both chains the deployed contracts paid a captured vote in full after the resolver's call. No rule that counts only votes protects a market beyond what a visible manipulation would cost the stake that votes. Past captures that took nothing from the protocol left the token within its ordinary noise, and even a loss of 14% would leave large markets beyond the limit; the oracle's own penalty protects a few thousand dollars and nothing against a coalition that can block. The margin is a second weakness. It has been one voter wide on twelve contested votes that decided a payout, and raising the line moved fifteen points of weakness from capture to blocking. Halting trading in a market while its question is before the vote caps the prize at its size when the vote was called. The vote settles every question disputed twice on the largest prediction market, on the strength of a safety argument whose numbers were never checked against the votes it produced.

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Appendix A. Proofs

Shares are of the staked supply S, normalized to one; the rule, the blocks σ, h_F, h_T, m and the tallies $x_F(y), x_T(y)$ are those of Section 5. The artifact holds the proofs of Propositions 1, 2 and 3 (direct readings of the rule), of Theorem 2 and Proposition 4 (monotonicity arguments), and the numerical check of the noisy coordination game.

Proof of Theorem 1: For the guarantee we choose the response of the remaining stake that is worst for F. Monotonicity of the rule in x_F says that worst case is $y = 0$ with the whole remainder voting T, so $x_F = \sigma + h_F$ and F resolves exactly when $\sigma + h_F > \theta$. For the impossibility we bound x_F from above: when $h_T \geq 1 - \theta$, every y gives $x_F(y) \leq \sigma + h_F + m = 1 - h_T \leq \theta$. For the equilibria, suppose every remaining voter votes F; then $x_F = 1 - h_T > \theta$ and F resolves. A remaining voter of stake $s \leq \theta - h_T$ that votes T or abstains leaves T at most $h_T + s \leq \theta$, so T cannot resolve: either F still resolves and the deviator pays λ instead of earning the pool rate, or the request rolls and it earns nothing. Nobody deviates. We need the bound on s, because the pool rate $\lambda(1 - v)/v$ of a winning side v falls with v: a voter with s between $\theta - h_T$ and $1 - 2h_T$ that votes T makes T resolve and earns $\lambda(1 - h_T - s)/(h_T + s)$, more than the $\lambda h_T/(1 - h_T)$ it earns with F. The symmetric argument, with $s \leq \theta - \sigma - h_F$, gives the T equilibrium whenever

$$1 - \sigma - h_F \ > \ \theta.$$

□

Proof of Theorem 3: We use the full-participation line $x + y = 1$ as the reference. Neutrality and monotonicity make $\{x : f(x, 1-x) = F\}$ an up-set and $\{x : f(x, 1-x) = T\}$ a down-set that mirror each other, so on the line F resolves for $x > \theta_f$ and not for $x < \theta_f$, T for $x < 1 - \theta_f$ and not for $x > 1 - \theta_f$, and nothing strictly between; this gives $D = 2\theta_f - 1$ and forces $\theta_f \geq 1/2$.

(a) The honest rest votes T, which is also the response worst for a coalition voting F: any abstention or F vote by the rest would raise x or lower y, and monotonicity would keep F resolved. Voting F puts the tally on the line at $x = \sigma + h_F$, and any other split of the coalition's stake lowers x or raises y; so F resolves for $\sigma + h_F > \theta_f$ and under no split for $\sigma + h_F < \theta_f$, which gives the infimum.

(b) To block T the coalition gives it no votes, and among the remaining splits voting F is best: every split leaves T's tally at $1 - \sigma - h_F$, and voting F maximizes x, so by monotonicity it blocks whenever any split does. Voting F puts the tally on the line at $x = \sigma + h_F$, where T resolves for $\sigma + h_F < 1 - \theta_f$ and not for $\sigma + h_F > 1 - \theta_f$; hence $\sigma_{\mathrm{blk}}(h_F) = 1 - \theta_f - h_F$. When abstention counts as opposition, abstaining and voting F present the rule with the same case against T, so abstaining blocks whenever voting F does. When the rule depends only on $x/(x + y)$ and $\theta_f < 1$, abstaining leaves T with $(1 - \sigma - h_F)/(1 - \sigma)$ of the votes cast, below θ_f only for $\sigma > 1 - h_F/(1 - \theta_f)$, a larger requirement than $1 - \theta_f - h_F$. Adding (a) and (b),

$$\sigma_{\mathrm{cap}}(h_F) + \sigma_{\mathrm{blk}}(h_F) = 1 - 2h_F.$$

□

Sufficiency of (1): We show that the cost of a capturing coalition of share $\sigma > \theta - h_F$ minus its largest gain does not fall in σ; it is then positive for every such coalition once it is positive in the limit $\sigma \downarrow \theta - h_F$, which is (1). The gain is $(1 - \bar{p})Q_{\max}$, which does not depend on σ, plus the pool transfer $\lambda V \sigma(1-v)/v$ of Proposition 2, largest when the remaining stake votes T and the winning stake is $v = \sigma + h_F$; so the largest gain is $(1 - \bar{p})Q_{\max} + \lambda V g(\sigma)$ with $g(\sigma) = \sigma(1 - \sigma - h_F)/(\sigma + h_F)$. The cost rises at rate $C'(\sigma) \geq \lambda V(1-\theta)/\theta$; when every staked token loses a common fraction δ of its value, $C(\sigma) = \delta\sigma V$ and the rate is δV. The pool share rises at most as fast, since with $v = \sigma + h_F \geq \theta \geq h_F$,

$$g'(\sigma) \ = \ \frac{h_F - v^2}{v^2} \ \leq \ \frac{h_F}{\theta^2} - 1 \ \leq \ \frac{1 - \theta}{\theta}.$$

□

Proof of Theorem 4: With two answers a No share is paid $1 - \rho(f) = \rho(1 - f)$, so each answer is paid ρ of its own share, and we write f for the share of F. A vote for "too early" that does not lead the tally lowers the share of Yes exactly as a vote for No does, so for (i) we count it as one. We first locate the share f of the votes cast that F can reach. With the coalition's stake split into a for F, b for T and c abstaining, and some vote cast, $f = (h_F + a)/(h_F + h_T + a + b)$. Voting T with everything gives $f = h_F/\pi$ and voting F with everything gives $f = (h_F + \sigma)/\pi$, and every split lies between, since $a + b \leq \sigma$ gives

$$\frac{h_F}{\pi} \leq \frac{h_F + a}{h_F + h_T + a + b} \leq \frac{h_F + a}{h_F + h_T + a} \leq \frac{h_F + \sigma}{\pi},$$

the last step because $(h_F + x)/(h_F + h_T + x)$ rises in x. As ρ is continuous and nondecreasing with slope at most $1/(1-2\mu)$, (i) follows, and the DVM's own resolution never enters, so a roll changes nothing. Part (ii) is $\rho((h_F+\sigma)/\pi) = 1$ for the largest share F can reach, that is $h_F + \sigma \geq (1-\mu)\pi$.

For (iii), let P be a monotone neutral rule with $P(f) = 1$ for $f \geq 1 - \mu$. Neutrality gives $P(\mu) = 1 - P(1 - \mu) = 0$, so the honest electorate $h_F = h_T = \mu\pi$ and the coalition $\sigma = (1 - 2\mu)\pi$ move F's share from μ to $1 - \mu$ and the payout by one, a shift of $1/(\pi(1 - 2\mu))$ per unit of stake. Small coalitions do no better against every electorate: for a coalition with share $s \leq \mu$ of the votes cast, $P(f+s) - P(f)$ integrates to s over the feasible baselines $f \in [\mu - s, 1 - \mu]$, since P is 0 up to μ and 1 from $1 - \mu$, so some honest electorate lets the coalition move the payout by at least $s/(1 - 2\mu + s)$. If P jumps by J at f_0, as a rule paying only zero or one must somewhere in $[\mu, 1 - \mu]$, an honest electorate that leaves F's share just below f_0 lets an arbitrarily small coalition produce the jump. For (iv), by (i) the coalition's votes move each of its Q_j shares in market

j by at most $\sigma/(\pi(1-2\mu))$, with Q_j at most that market's open interest OI_j, while its stake loses $\delta\sigma V$ once, however many markets the round settles. Here δ already counts the change in the coalition's penalty-pool payoff, which is at most λ/θ of its stake per request, since that payoff lies between $-\sigma\lambda$ on a losing side and $\sigma\lambda(1-\theta)/\theta$ on a side that resolves (Proposition 2). So when $\sum_j \mathrm{OI}_j \le \delta(1-2\mu)\pi V$,

$$\frac{\sigma\sum_j Q_j}{\pi(1-2\mu)} \le \frac{\sigma\sum_j \mathrm{OI}_j}{\pi(1-2\mu)} \le \delta\sigma V.$$

For (v), we give "too early" the coalition's whole stake, since a vote anywhere else lowers its tally or raises a rival's. It then holds σ of the votes against h_F for F and h_T for T, and it alone leads exactly when

$$\sigma > \max\{h_F, h_T\},$$

which at full participation, where $h_T = 1-\sigma-h_F$, is $\sigma > \max\{h_F, (1-h_F)/2\}$. □

Proof of Corollary 1: The second sentence is part (iv) of Theorem 4. For the first, we take the electorate of part (iii), $h_F = h_T = \mu\pi$, and a coalition of $(1-2\mu)\pi$ of the stake holding Q shares of F. Voting F instead of T moves the payout of F from $P(\mu) = 0$ to $P(1-\mu) = 1$, so the coalition gains Q and its stake loses $\delta(1-2\mu)\pi V$. The manipulation pays once Q, which can be as large as the open interest, exceeds

$$\delta(1-2\mu)\pi V.$$

□

Remark 1 (Paying for truthful votes). Freeman, Lahaie and Pennock build the cost into the mechanism: arbiters are paid from trading fees by a peer-prediction rule under which, when the others report truthfully, a truthful report earns $k\Delta/2$ more than a false one in expectation, with k the arbiter's payment scale and $\Delta \in (0,1]$ the update strength of the question [1]. To price it on a stake-weighted vote we stipulate the coalition form of their one-deviator lemma: a pool R divided among the voters in proportion to stake costs a coalition with share σ/π of the votes cast $(\sigma/\pi)R\Delta/2$ in expectation when it votes F. It gains at most $Q\sigma/(\pi(1-2\mu))$ by Theorem 4(i), so $R \ge 2Q/(\Delta(1-2\mu))$ deters every coalition on a market in which one can hold Q shares, and no smaller pool does, as the electorate of Theorem 4(iii) shows. Before the three wallets revealed in March 2025, a coalition could have bought 249,274 Yes shares (measured as in Section 5.7); at $\mu = 0.2$, deterring every coalition would have taken a pool of at least \$831,000, against the \$15,000 the penalty pool paid.

Remark 2 (The oracle's own penalty). A coalition that votes F when T resolves pays $\lambda = 0.1\%$ of its stake and forgoes the pool rate $\lambda(1-v)/v$ of a vote for T (Proposition 2, $v > \theta$): it loses between λ and λ/θ relative to voting T, the limit's δ above. Against the honest electorate of Theorem 2, a coalition holding $1-\theta-h_F$ pays nothing at all. Since R_μ fixes the payout on the first round's tally, it votes F there to keep T below the line, lets the request roll, and votes T in the next round; no penalty is charged in a round that rolls. At $\theta = 0.65$ and $h_F = 0.1$ that is a quarter of the stake, less than today's four largest stakers hold.

Remark 3 (When the rest may vote the blocker's answer). If the electorate outside the coalition may vote any way, including for the coalition's own answer, a coalition that wants no answer to resolve must block both. A T-vote of more than $1-\theta_f$ vetoes F in every completion, and by neutrality an F-vote of more than $1-\theta_f$ vetoes T, so any coalition larger than $2(1-\theta_f)$ suffices; below $1-\theta_f$ the rest can resolve F by voting it unanimously. Under the staked-supply rule $1-\theta_f$ abstaining already blocks both. Under the votes-cast rule, with the coalition's split (a,b,c) for F, T and abstention, blocking F when the rest votes F needs $1-\sigma+a \le \theta_f(1-\sigma+a+b)$ and blocking T when the rest votes T needs $1-\sigma+b \le \theta_f(1-\sigma+a+b)$; adding them gives $2(1-\theta_f)(1-\sigma) \le (2\theta_f-1)(a+b) \le (2\theta_f-1)\sigma$, so the votes-cast rule needs $2(1-\theta_f)$ as well. At full participation this is the only sense in which the denominator changes a blocking requirement, and it requires voters outside the coalition to vote for the answer the coalition wants.

Appendix B. A market-referenced fallback

An appealing alternative to tally settlement pays a contested vote at the market's own pre-dispute price.

Definition 2 (F_μ). Fix $\mu \in (0, 1-\theta)$ and a reference price $\hat{p}$, the time-weighted price of an F share over a window that ends before the question can be disputed. A resolution pays as the oracle says, one and zero or a half to each side for the 0.5 answer, when the winning answer holds at least $1-\mu$ of the votes cast or the question resolved without a vote; otherwise, and for a request the oracle deleted, every share is paid its reference price, $\hat{p}$ to F and $1-\hat{p}$ to T.

The margin is measured against the votes cast; against the staked supply an abstaining coalition of 16.7% could trigger the fallback at $\mu = 0.2$.

Proposition 4 (What the fallback guarantees). Under F_μ with $\hat{p} < 1$, against an honest electorate with full participation and a block $h_F \le \mu$ for the coalition's answer F: (i) the coalition can make F shares pay one dollar if and only if $\sigma \ge 1-\mu-h_F$; (ii) it can make the question pay the reference price if and only if $\sigma > \mu - h_F$; (iii) the two requirements add up to $1-2h_F$, as in Theorem 3; (iv) a coalition below $1-\mu-h_F$ changes no share's payout by more than $\hat{p}$, and on F shares bought at $\bar{p}$ it gains at most $\hat{p}-\bar{p}$ per share.

On the March question, with a pre-dispute price of 0.17 for Yes, Yes holders would have received 0.17 a share. Forcing the fallback takes only $\mu - h_F$: with no honest support,

today's two largest stakers can do it at $\mu = 0.2$, and holders of the answer an honest vote would reject can use it as insurance. The reference price must be the market's own, since with a constant such as one half forcing the fallback on a market priced at 17 cents is a sure profit; Proposition 5 shows that the market's own price fails too.

A coalition that can force the fallback can also set the price it is paid.

Proposition 5 (Buying the reference price). Let $\hat{p}$ be the time average, over the reference window, of the price of an F share, where the price at each moment is the last trade or the midpoint of the best bid and ask on a limit order book that fills incoming orders against the best resting orders at their prices and, when the midpoint is used, always has an ask. Suppose a coalition keeps a bid for F at $\bar{p}$ throughout the window and buys every share offered at or below $\bar{p}$; if the price is the last trade, it also buys one share at $\bar{p}$ or above just before the window opens. Then $\hat{p} \geq \bar{p}$, the fallback pays each share its bid bought in the window at least what it cost, and it pays every share of F held before the window the price $\hat{p}$.

Proof: While the bid stands, a seller trades with it at $\bar{p}$ or with a higher bid, and a buyer pays the best ask, which exceeds $\bar{p}$ because an ask at or below the standing bid would have filled; so every trade in the window is at $\bar{p}$ or above, and the share bought before the window puts the last trade there when the window opens. The best bid is at least $\bar{p}$ and the best ask exceeds it, so the midpoint exceeds $\bar{p}$ too. Either price stays at or above $\bar{p}$ throughout the window, and so does its time average, $\hat{p} \geq \bar{p}$. The bid is filled as a resting order at its own price, so its q fills in the window cost $q\bar{p}$, which the fallback repays with $q\hat{p} \geq q\bar{p}$. □

The push costs nothing at settlement, but it needs cash: the coalition pays $q\bar{p}$ during the window and recovers it when the question settles. The shares outstanding do not bound q, since anyone can split collateral into sets and sell the F half into the bid, so the coalition's cash limits the push, together with the stake it needs to force the fallback (Proposition 4). Paying the reference price only when a contested vote goes against the market's favorite fares no better, since the push carries F above one half. This is why the rule of Section 6 reads only the votes.

Appendix C.
Tally settlement on the oracle's history

Figure 5 draws tally settlement against the deployed rule, and Table 3 gives the replay of Section 6.3 for six values of μ. Questions answered one half are paid within 0.3 cents of a half at the median and 18 cents at most, because a vote for one half counts half.

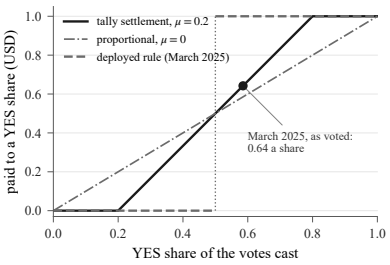


Figure 5. What a disputed question pays a Yes share as a function of Yes's share of the votes cast, for two answers and full participation. The deployed rule (threshold one half, as in March 2025) pays nothing or a dollar; with a third answer or abstentions, a vote that carries no answer past the line rolls and settles nothing. Proportional settlement [1] is the diagonal, and tally settlement with $\mu = 0.2$ the ramp. The circle is the March vote as cast.

TABLE 3. TALLY SETTLEMENT REPLAYED OVER EVERY POLYMARKET VOTE THAT DECIDED A PAYOUT, ON THE FIRST ROUND'S TALLY; $\mu = 0$ IS PROPORTIONAL SETTLEMENT. "BELOW FULL" IS THE SHARE OF THE 457 YES-OR-NO RESOLUTIONS THAT BOTH RULES PAY (354 IN THE 65% ERA) WHOSE WINNER IS PAID LESS THAN A DOLLAR, AND "CUT" THE MEDIAN SHORTFALL IN CENTS. "P10" IS THE TENTH PERCENTILE OF THE WINNER'S PAYOUT IN THE 36 CONTESTED YES-OR-NO RESOLUTIONS, "10%" THE MOST A TENTH OF THE SUPPLY MOVES ANY PAYOUT IN CENTS AT FULL PARTICIPATION, AND "MARCH" THE PAYOUT OF THE MARCH 2025 "YES".

μ	below full	65% era	cut	P10	10%	March
0	72.6%	70.3%	0.3	0.65	10.0	0.59
0.10	7.0%	5.6%	8.4	0.68	12.5	0.61
0.15	3.9%	3.7%	24.4	0.71	14.3	0.62
0.20	3.3%	2.8%	20.5	0.74	16.7	0.64
0.25	3.1%	2.5%	15.1	0.79	20.0	0.67
0.30	2.4%	2.3%	7.8	0.86	25.0	0.71

Appendix D.
The March 2025 request, the slashing port and the counterfactuals

The port reproduces `VotingV2`'s resolution test (total votes above the participation floor and modal votes above θ times the frozen supply) and its penalty arithmetic: the pool is $\lfloor(\lambda_n(S-v)+\lambda_w(v-v^*))/10^{18}\rfloor$, a wrong or absent token pays $\lceil s\lambda/10^{18}\rceil$, and a correct token receives $\lfloor s \cdot \text{pool}/v^* \rfloor$; the ceiling on penalties is the contract's `Math.ceilDiv`, found when a floor-based first version disagreed with the chain by one wei on half the voters. Against the 819 penalty events emitted so far for each of the two March requests, every voter with an event matches to the wei (396 and 397 voters, 0 mismatches); the remaining events belong to absent stakers whose round stake is not in the vote events. The same arithmetic gives the counterfactuals in the lower part of Table 4.

Appendix E.
Captures of on-chain votes

For each capture we compute the governance token's abnormal log return against ETH over the week that starts

TABLE 4. THE REQUEST THAT DECIDED THE MINERALS-DEAL MARKET'S PAYOUT (ROUND 10085, SECOND REQUEST), DECODED FROM THE VOTING CONTRACT, AND THE COUNTERFACTUALS OF THE SAME ARITHMETIC (UMA).

Staked supply S (UMA)	22,845,749
Agreement requirement, $\theta = 50\%$ of S	11,422,874
"Yes"	12,292,521 (53.8%)
"Too early"	8,706,371 (38.1%)
"No"	1,144
Did not vote	1,845,686 (8.1%)
Voters	514
Three wallets, revealed 22:58 to 23:00 UTC	5,083,491 (22.25%)
Margin of "Yes" over the requirement	869,647 (3.81%)
Stake the three wallets needed	4,213,845
Penalty pool, paid to "Yes" voters	10,553
of which to the three wallets	4,364
Open interest when the round ended (USDC)	987,056
"Yes" shares redeemed within a week (USDC)	1,129,633
Pre-dispute 24-hour mean price of "Yes"	0.17
Stake they would have needed at $\theta = 65\%$	7,640,707
shortfall at 65%	2,557,216
Needed with the votes-cast denominator, 50%	3,291,002
Stall coalition, 65% of staked	7,996,012
Stall coalition, 65% of votes cast, rest voting its answer (Remark 3)	15,992,024

TABLE 5. CAPTURES AND ATTEMPTED CAPTURES OF ON-CHAIN VOTES. WEEK: THE GOVERNANCE TOKEN'S SIMPLE ABNORMAL RETURN AGAINST ETH IN THE WEEK FROM THE CAPTURE; z: THE LOG RETURN OVER ITS STANDARD DEVIATION IN THE PRECEDING 365 DAYS; TOOK: WHETHER THE CAPTURE DRAINED OR DILUTED THE PROTOCOL. *ONE DAY: CURIO'S TOKEN WAS REPLACED THE NEXT DAY.

Capture	Token	Month	Took	Week	z
Build Finance	BUILD	2022-02	yes	-99.7%	-17.8
Audius	AUDIO	2022-07	yes	-16.0%	-1.0
Mango Markets	MNGO	2022-10	yes	-41.2%	-4.7
Tornado Cash	TORN	2023-05	yes	-34.5%	-2.4
Synthetify	SNY	2023-10	yes	-7.1%	-0.3
Curio	CGT	2024-03	yes	$-11.8\%^*$	-0.6
Steem	STEEM	2020-03	no	$+19.8\%$	$+1.9$
Maker flash vote	MKR	2020-10	no	-4.7%	-0.5
Curve, Mochi	CRV	2021-11	no	$+15.6\%$	$+0.7$
Compound TUSD	COMP	2022-02	no	-6.8%	-0.6
Yam	YAM	2022-07	no	-14.0%	-1.1
Indexed	NDX	2023-11	no	$+204.1\%$	$+3.9$
Compound 289	COMP	2024-07	no	-3.8%	-0.5
UMA	UMA	2025-03	no	-5.5%	-0.5

TABLE 6. RESOLUTION DESIGNS AND THEIR POSITION RELATIVE TO THE CLASS OF TALLY-ONLY RULES.

Design	Rule	In the class?
UMA DVM 2.0	modal answer $> 65\%$ of staked, else roll; deleted after 4 rolls	yes; block $35\% - h_F$, by abstaining
same, votes cast	modal answer $> 65\%$ of votes cast	yes; block $35\% - h_F$, by voting
Kleros	plurality of a stake-drawn jury; appeals with $2n+1$ jurors; incoherent jurors pay coherent ones	one round yes ($\theta_f = 1/2$); the appeal ladder no
Augur v2	escalating REP bonds; a bond of 2.5% of REP forks the token	no: bonds, not tallies; the terminal is a fork
Reality.eth	doubling bonds, timeout, designated arbitrator	no: bonds and an arbitrator
Bittensor Yuma	stake-weighted median of validator weights	yes; no roll; $\theta_f = 1/2$

on the day of the capture, from daily prices [45], and scale it by the standard deviation of the token's overlapping weekly abnormal returns over the 365 days before. Beanstalk and Venus's VIP-42 lack a year of prices and are left out. Pooled, the captures that took nothing from their protocol show no significant mean loss; the one-sided 95% bound on their mean weekly loss is 13%, or 9% without Indexed, whose token tripled against ETH that week. On each token's pre-event noise, a two-sided 5% test detects a mean loss of 14% four times in five. At the first group's median loss of 25%, 13 of the 51 contested markets remain beyond the rule's guarantee. Only one of the captures is of UMA's own vote; the others come from different protocols, successful and stopped captures are pooled, and a week's return against ETH does not isolate other news. For that one, UMA rose 0.8% in the day after the round closed while ETH fell 0.7%, outperformed ETH by 1.0 points over three days, and over the week fell 17.2% against ETH's 12.5%, 5.5 points in log returns, less than half the standard deviation of its weekly return relative to ETH over the preceding year (12.5 points); against BTC the weekly shortfall is 13.0 points with a standard deviation of 13.5.

Appendix F. Five resolution designs

Table 6 places the deployments discussed in Section 9 in the frame of Theorem 3.

Appendix G. Gas of the prototype

TallyRootTunnel and *TallyChildTunnel* carry the share of Section 8 as one more state-sync message through the FxPortal interfaces that UMA's oracle tunnel already uses; six tests run the path end to end with a mock bridge and check who may send; Table 7 lists the gas of each call. The relay's figure includes the mock bridge delivering to the child in the same transaction, which the real state sync does asynchronously, and the mock-based figure for the votes-cast reader omits the cold-storage reads of the live contract. The artifact also contains the market-referenced adapter of Appendix B, whose price accumulator costs 54,246 gas on every fill.

TABLE 7. GAS OF THE PROTOTYPE (FOUNDRY 1.8.5, SOLC 0.8.24, OPTIMIZER 200 RUNS).

Call	Gas	When
register (MarginReader, mainnet fork)	102,204	per disputed request
register (votes cast, mock)	61,304	per disputed request
settle, undisputed (TallyAdapter)	129,404	per question
settle, paid in full	135,819	per question
settle, paid by the ramp	155,719	per question
relay (TallyRootTunnel, mock bridge)	88,434	per disputed request
receive (TallyChildTunnel)	24,528	per disputed request